TEJ3M Robotics Assignment



Electronics

Instructions:

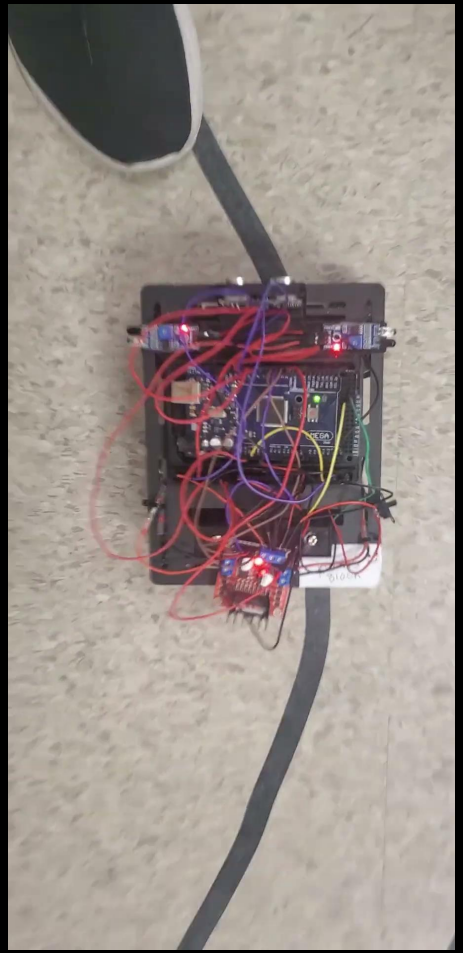
Using the robot you built and tested you will further modify its circuits and write the necessary code to get it to complete the tasks on the following slides. In each slide you will embed a short video of it completing the specified task. In the slide following the embedded video include a screenshot fo your code. Each task is worth 20 marks. 10 TI for successfully.accurate completion of the task and 10 C for how well you present the task in your video and how well you write/document/format your code.



Electronics

Task #1

Have the robot continuously move in a triangular path that has an approximate diameter of one foot.



Task #1

```
/*
  Shoaib Mahmood, Lucas Ha
  12/15/2025
  robot triangle
  this program will make our robot drive in a triangle around 1ft long
*/



int leftBwd = 2;
int leftFwd = 3;
int  rightBwd = 4;
int rightFwd = 5;
int vardelay = 1500;


void forward()
{
  digitalWrite(leftFwd,HIGH);
  digitalWrite(rightFwd,HIGH);
  delay(vardelay);
  digitalWrite(leftFwd,LOW);
  digitalWrite(rightFwd,LOW);
}

void backward()
{
  digitalWrite(leftBwd,HIGH);
  digitalWrite(rightBwd,HIGH);
}

void setup()
{

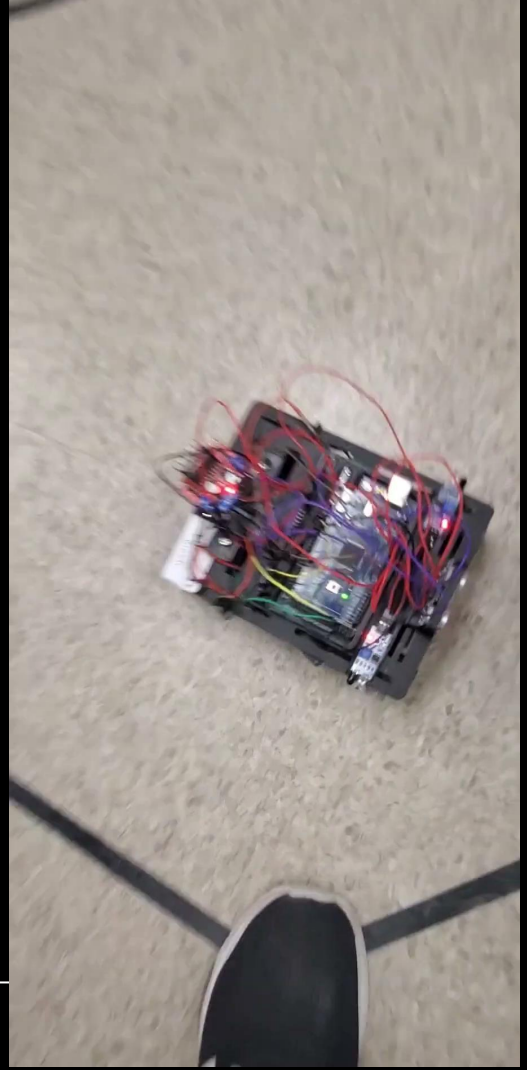
    Serial.begin(9600);
    pinMode(leftBwd,OUTPUT);
    pinMode(leftFwd,OUTPUT);
    pinMode(rightBwd, OUTPUT);
    pinMode(rightFwd,OUTPUT);

}
```

```
void loop()
{
    forward();
    digitalWrite(leftBwd,HIGH);
    digitalWrite(rightFwd,HIGH);
    delay(1275);
    digitalWrite(leftBwd,LOW);
    digitalWrite(rightFwd,LOW);
}
```

Task #2

Have the robot move in a square path whose side lengths are approximately one feet. When the robot reaches the end, it pauses one second and then follows in reverse the same path and stops at the start.



Task #2

```
/*
  Shoaib Mahmood, Lucas Ha
  12/15/2025
  robot square
  this program will make our robot drive in a square around 1ft wide
*/



int leftBwd = 2;
int leftFwd = 3;
int  rightBwd = 4;
int rightFwd = 5;
int fwdelay = 1500;
int bwdelay = 1500;


void forward()
{
  digitalWrite(leftFwd,HIGH);
  digitalWrite(rightFwd,HIGH);
  delay(fwdelay);
  digitalWrite(leftFwd,LOW);
  digitalWrite(rightFwd,LOW);
}

void backward()
{
  digitalWrite(leftBwd,HIGH);
  digitalWrite(rightBwd,HIGH);
  delay(bwdelay);
  digitalWrite(leftBwd,LOW);
  digitalWrite(rightBwd,LOW);
}

void setup()
{

    Serial.begin(9600);
    pinMode(leftBwd,OUTPUT);
    pinMode(leftFwd,OUTPUT);
    pinMode(rightBwd, OUTPUT);
    pinMode(rightFwd,OUTPUT);

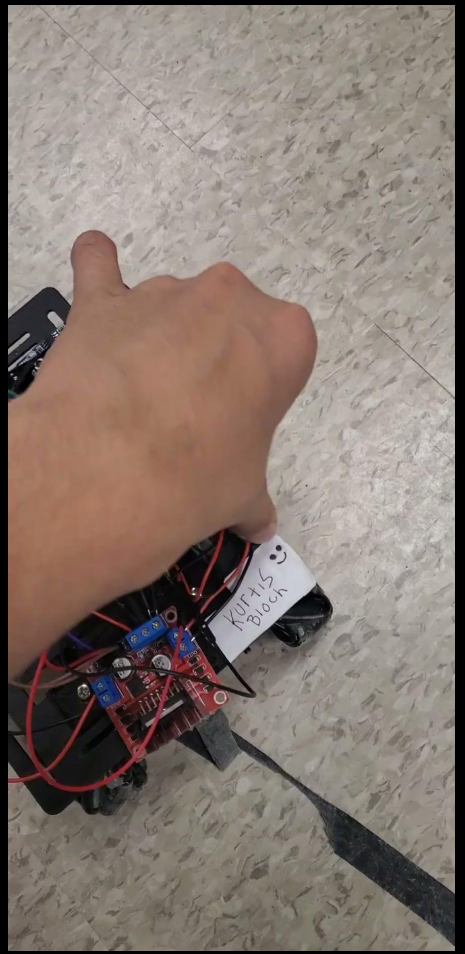
}
```

```
void loop()
{
    for(int i=0;i<4;i++)
    {
      digitalWrite(leftBwd,HIGH);
      digitalWrite(rightFwd,HIGH);
      delay(941);
      digitalWrite(leftBwd,LOW);
      digitalWrite(rightFwd,LOW);
      forward();
    }

    for(int i=0;i<4;i++)
    {
      backward();
      digitalWrite(leftFwd,HIGH);
      digitalWrite(rightBwd,HIGH);
      delay(941);
      digitalWrite(leftFwd,LOW);
      digitalWrite(rightBwd,LOW);
    }
}
```

Task #3

Add in the piezo buzzer and PCB RGB components to your robot circuit. Make your robot move straight continuously and while doing so display the green LED. Moving your hand in front of the sonar sensor stops the robot, turns the RGB red, forces the piezo buzzer to beep for one second and then moves the robot in reverse for one second while lighting up the blue RGB. At this point it turns a random degree, stops and continues forward with the green RGB.



Task #3

```
/*
  Shoaib Mahmood, Lucas Ha
  12/15/2025
  task 3 Sonar sensor
  this program will demonstrate the use of the Sonar sensor, RGB LED, and Piezzo buzzer
*/



int leftFwd = 2;
int leftBwd = 3;
int  rightFwd = 4;
int rightBwd = 5;

int buzzer = 50;
int red = 35;
int green =41;
int blue = 39;



int fwdelay = 1500;
int bwdelay = 1000;

const int trigpin = 9;
const int echopin = 10;
long duration;
int distance;

int randomDegree = 0;

void forward()
{
  digitalWrite(leftFwd,HIGH);
  digitalWrite(rightFwd,HIGH);
  delay(fwdelay);
  digitalWrite(leftFwd,LOW);
  digitalWrite(rightFwd,LOW);
}
void backward()
{
  digitalWrite(leftBwd,HIGH);
  digitalWrite(rightBwd,HIGH);
  delay(bwdelay);
  digitalWrite(leftBwd,LOW);
  digitalWrite(rightBwd,LOW);
}
```

```
void setup()
{

    Serial.begin(9600);
    pinMode(leftBwd,OUTPUT);
    pinMode(leftFwd,OUTPUT);
    pinMode(rightBwd, OUTPUT);
    pinMode(rightFwd,OUTPUT);
    pinMode(red,OUTPUT);
    pinMode(green,OUTPUT);
    pinMode(blue,OUTPUT);
    pinMode(trigpin,OUTPUT);
    pinMode(echopin, INPUT);
}
```

```
void loop()
{
    randomSeed(analogRead(0));
    digitalWrite(green,HIGH);
    digitalWrite(leftFwd,HIGH);
    digitalWrite(rightFwd,HIGH);


    digitalWrite(trigpin,LOW);
    delayMicroseconds(2);
    digitalWrite(trigpin,HIGH);
    delayMicroseconds(10);
    digitalWrite(trigpin,LOW);

    duration = pulseIn(echopin,HIGH);
    distance = duration*0.034/2;
    Serial.println(distance);

    if(distance<=5 && distance >0)
    {
      digitalWrite(leftFwd,LOW);
      digitalWrite(rightFwd,LOW);
      digitalWrite(green,LOW);
      digitalWrite(red,HIGH);
      tone(buzzer, 2500, 1000);
      delay(1000);
      digitalWrite(red,LOW);
      digitalWrite(blue,HIGH);
      backward();
      digitalWrite(blue,LOW);

      randomDegree = random(10,3800);

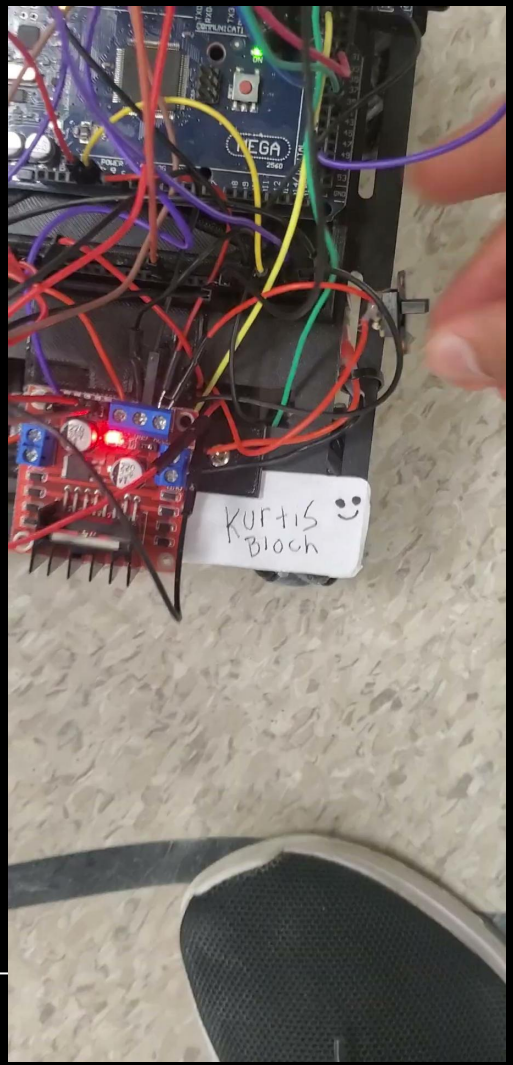
      digitalWrite(rightFwd,HIGH);
      digitalWrite(leftBwd,HIGH);
      delay(randomDegree);
      digitalWrite(rightFwd,LOW);
      digitalWrite(leftBwd,LOW);



    }
```

Task #4

Using black electrical tape create any shape you want on the floor. Make sure its large enough to give your robot room to move around in it. Make sure robot continuously move inside the shape but all the while making sure it nevers leaves its boundaries. Every time it detects a boundary it needs to back up and turn in a random direction before proceeding again. While the robot is moving forward flash the RGB. When it detects a boundary play a tone on the piezo.



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Task #4

```
/*
  Shoaib Mahmood, Lucas Ha
  12/15/2025
  task 4 floor IR sensor
  this program will make the robot move while staying inside a region outlined by black tape.
  The floor IR sensors on the robot will stop it from going outside the taped region
*/



int leftFwd = 2;
int leftBwd = 3;
int  rightFwd = 4;
int rightBwd = 5;

int buzzer = 50;
int red = 35;
int green =41;
int blue = 39;

int fwdelay = 1500;
int bwdelay = 1000;

const int trigpin = 9;
const int echopin = 10;
long duration;
int distance;


int leftIR = 11;
int rightIR = 12;
int floor1 = 22;
int floor2 = 26;

int floor1val = 67;
int floor2val = 69;

int turnDelay = 0;

void forward()
{
  digitalWrite(leftFwd,HIGH);
  digitalWrite(rightFwd,HIGH);
  delay(fwdelay);
  digitalWrite(leftFwd,LOW);
  digitalWrite(rightFwd,LOW);
}
```

```
void backward()
{
  digitalWrite(leftBwd,HIGH);
  digitalWrite(rightBwd,HIGH);
  delay(bwdelay);
  digitalWrite(leftBwd,LOW);
  digitalWrite(rightBwd,LOW);
}


void distanceCheck()
{
    digitalWrite(trigpin,LOW);
    delayMicroseconds(2);
    digitalWrite(trigpin,HIGH);
    delayMicroseconds(10);
    digitalWrite(trigpin,LOW);

    duration = pulseIn(echopin,HIGH);
    distance = duration*0.034/2;
    Serial.println(distance);
}

void turn()
{
  digitalWrite(rightFwd,HIGH);
  digitalWrite(leftBwd,HIGH);
  delay(turnDelay);
  digitalWrite(rightFwd,LOW);
  digitalWrite(leftBwd,LOW);
}
```

```
void setup()
{

    Serial.begin(9600);
    pinMode(leftBwd,OUTPUT);
    pinMode(leftFwd,OUTPUT);
    pinMode(rightBwd, OUTPUT);
    pinMode(rightFwd,OUTPUT);
    pinMode(red,OUTPUT);
    pinMode(green,OUTPUT);
    pinMode(blue,OUTPUT);

    pinMode(trigpin,OUTPUT);
    pinMode(echopin, INPUT);

    pinMode(leftIR, INPUT);
    pinMode(rightIR,INPUT);
    pinMode(floor1, INPUT);
    pinMode(floor2,INPUT);
}


void loop()
{
   randomSeed(analogRead(0));
   digitalWrite(leftFwd,HIGH);
   digitalWrite(rightFwd,HIGH);
   digitalWrite(blue,HIGH);
   floor1val = digitalRead(floor1);
   floor2val = digitalRead(floor2);
   Serial.println(floor1val);
   if(floor1val == 0)
   {
      digitalWrite(leftFwd,LOW);
      digitalWrite(rightFwd,LOW);
      digitalWrite(blue,LOW);
      tone(buzzer,2500,1000);
      delay(1000);
      bwdelay = 800;
      backward();
      turnDelay = random(10,3800);
      turn();
   }

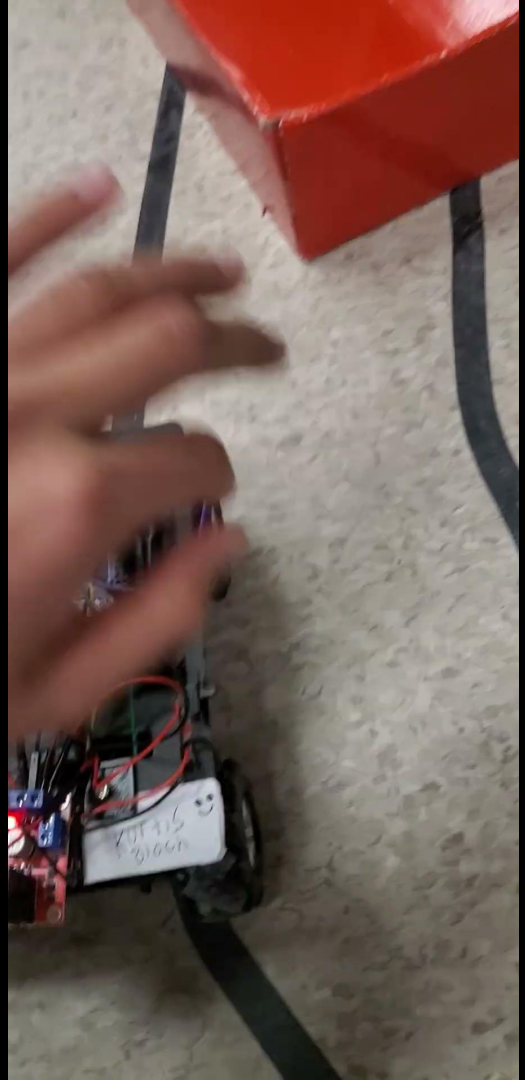
}
```

Task #5

Describe a task below you come up with that your robot will complete that at a minimum uses the side IR sensors .

Description:

Place the robot in a hallway or near a long wall. As it moves forward, it uses left and right IR sensors to maintain a steady distance from the wall. If the left sensor detects the wall is too close, the robot veers right; if the right sensor is too close, it veers left. Additionally, when the robot is too close to a wall, it outputs a red LED while playing a tone for 1 second, before correcting itself. While correctly following the wall, a green RGB LED is displayed.



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Task #5

```
/*
  Shoaib Mahmood, Lucas Ha, Victor
  12/15/2025
  task 5 Side IR sensor
  This program will make the robot move between 2 walls.
  If the robot gets too close to a wall, the IR sensor
  will sense it and the robot will turn to correct itself.
*/



int leftFwd = 2;
int leftBwd = 3;
int  rightFwd = 4;
int rightBwd = 5;

int buzzer = 50;
int red = 35;
int green =41;
int blue = 39;

int fwdelay = 1500;
int bwdelay = 1000;

const int trigpin = 9;
const int echopin = 10;
long duration;
int distance;


int leftIR = 11;
int rightIR = 12;

int leftIRval;
int rightIRval;

int turnDelay = 200;

void forward()
{
  digitalWrite(leftFwd,HIGH);
  digitalWrite(rightFwd,HIGH);
  delay(fwdelay);
  digitalWrite(leftFwd,LOW);
  digitalWrite(rightFwd,LOW);
}
```

```
void backward()
{
  digitalWrite(leftBwd,HIGH);
  digitalWrite(rightBwd,HIGH);
  delay(bwdelay);
  digitalWrite(leftBwd,LOW);
  digitalWrite(rightBwd,LOW);
}


void turnRight()
{
  digitalWrite(rightFwd,HIGH);
  digitalWrite(leftBwd,HIGH);
  delay(turnDelay);
  digitalWrite(rightFwd,LOW);
  digitalWrite(leftBwd,LOW);
}

void turnLeft()
{
  digitalWrite(rightBwd,HIGH);
  digitalWrite(leftFwd,HIGH);
  delay(turnDelay);
  digitalWrite(rightBwd,LOW);
  digitalWrite(leftFwd,LOW);
}

void setup()
{
    Serial.begin(9600);
    pinMode(leftBwd,OUTPUT);
    pinMode(leftFwd,OUTPUT);
    pinMode(rightBwd, OUTPUT);
    pinMode(rightFwd,OUTPUT);
    pinMode(red,OUTPUT);
    pinMode(green,OUTPUT);
    pinMode(blue,OUTPUT);

    pinMode(trigpin,OUTPUT);
    pinMode(echopin, INPUT);

    pinMode(leftIR, INPUT);
    pinMode(rightIR,INPUT);
}
```

```
void loop()
{

   digitalWrite(leftFwd,HIGH);
   digitalWrite(rightFwd,HIGH);
   digitalWrite(green,HIGH);
   leftIRval = digitalRead(leftIR);
   rightIRval = digitalRead(rightIR);
   Serial.println(leftIRval);
   Serial.println(rightIRval);
   if(leftIRval == 0)
   {
     digitalWrite(leftFwd,LOW);
     digitalWrite(rightFwd,LOW);
     digitalWrite(green,LOW);
     digitalWrite(red,HIGH);
     tone(buzzer, 2500, 1000);
     delay(1000);
     turnRight();
     digitalWrite(red,LOW);
   }
   if(rightIRval == 0)
   {
     digitalWrite(leftFwd,LOW);
     digitalWrite(rightFwd,LOW);
     digitalWrite(green,LOW);
     digitalWrite(red,HIGH);
     tone(buzzer, 2500, 1000);
     delay(1000);
     turnLeft();
     digitalWrite(red,LOW);
   }



}
```

Marks TI 10 C 10

Task #1

Task #2

Task #3

Task #4

Task #5